

Real-Time Time-Dependent Density Functional Theory for Periodic and Spin-Polarized Systems with the Real-Space Multigrids (RMG)[†]

Jacek Jakowski,^{*,‡,¶} Wenchang Lu,[§] Liangbo Liang,[‡] Emil Briggs,[§] Jerzy Bernholc,[§] and Panchapakesan Ganesh[‡]

[‡]*Center For Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA*

[¶]*Computational Sciences and Engineering Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA*

[§]*Department of Physics, North Carolina State University, Raleigh, NC 27695-8202, USA*

E-mail: jakowskij@ornl.gov

Phone: +1 865 574 6174

Abstract

[Abstract: to be written last, after all sections are complete. Approximately 200 words. Should mirror paper 1 abstract structure: (1) what we present, (2) what problem it solves, (3) key methodological contribution, (4) validation/benchmarking, (5) scalability demonstration and outlook.]

1 Introduction

Real-time time-dependent density functional theory (RT-TDDFT) provides a direct first-principles route for following electronic response after an applied perturbation. By propagating the electronic degrees of freedom in time, RT-TDDFT gives access to optical response, charge-transfer dynamics, beam-induced electronic excitations, plasmonic excitation and energy transfer, nonlinear response, and ultrafast electron dynamics within a single initial-value framework.^{1–6} For extended materials, low-dimensional systems, magnetic compounds, and defects in solids, the same real-time perspective is attractive because the relevant response may depend simultaneously on crystal momentum, spin polarization, local defect states, and long-range periodic environment. These requirements motivate scalable RT-TDDFT implementations that can treat periodic boundary conditions and large supercells without abandoning a first-principles electronic-structure description.

[†]Notice of Copyright: This manuscript has been authored by UT-Battelle, LLC, under contract DE-AC05-00OR22725 with the US Department of Energy (DOE). The publisher acknowledges the US government license to provide public access under the DOE Public Access Plan (<http://energy.gov/downloads/doe-public-access-plan>).

Table 1: Summary of representative external-field perturbations relevant to time-dependent electronic-response calculations. The table is intended as a conceptual map. Among the perturbations summarized here, the present work treats the homogeneous periodic optical response.

Physical probe / response	Representative perturbation	Field character	Typical applications	Relation to this work
Homogeneous optical electric field	Length-gauge dipole coupling for finite systems; homogeneous $\mathbf{A}(t)$ velocity-gauge representation for periodic systems	Long-wavelength $\mathbf{q} \rightarrow 0$ optical limit	Optical response, ultrafast laser-driven electron dynamics, pump/probe-type response	Present work
Finite- $\mathbf{q}$ longitudinal perturbation	$\Phi_{\mathbf{q}}(t)e^{i\mathbf{q}\cdot\mathbf{r}}$	Longitudinal electric field / density response	Finite- $\mathbf{q}$ density response, EELS/loss spectra, plasmon dispersion	Broader perspective; not implemented here
Finite- $\mathbf{q}$ transverse electromagnetic perturbation	$\mathbf{A}_{\mathbf{q}}(t)e^{i\mathbf{q}\cdot\mathbf{r}}$, with $\mathbf{q} \cdot \mathbf{A}_{\mathbf{q}} = 0$ in Coulomb gauge	Transverse electromagnetic response with associated magnetic field	Finite- $\mathbf{q}$ transverse optical or radiative response beyond the homogeneous optical limit	Broader perspective; not implemented here
Magnetic / spin perturbation	Time-dependent magnetic field, Zeeman coupling, or spin-dependent perturbation	Magnetic or spin response	Spin dynamics, magnetic resonance, magnetic excitations, magnons in appropriate noncollinear spin-dynamical frameworks	Not presently implemented as such
Multipolar / chiroptical perturbations	Electric quadrupole, magnetic dipole, field gradients, or mixed electric/magnetic response	Beyond electric-dipole response	Beyond-dipole spectroscopy, ECD, MCD	Broader perspective; not implemented here

Periodic systems require special care in the representation of a homogeneous optical perturbation. In finite systems, the electric-dipole interaction is often written in the length gauge through the position operator. Under periodic boundary conditions, however, the ordinary position operator is not a lattice-periodic observable, and the macroscopic polarization of an extended insulator is defined through Berry-phase or related periodic formulations rather than through a simple dipole moment.^{7–9} A natural periodic formulation of the long-wavelength optical response therefore uses a spatially homogeneous vector potential and evaluates the induced current response.^{10,11} The homogeneous vector potential represents the optical $\mathbf{q} \rightarrow 0$ limit; the restriction to vertical transitions comes from the long-wavelength approximation, not from the velocity gauge itself.

Table 1 places this homogeneous optical limit in the broader landscape of external-field representations used in time-dependent electronic-response theory. The table is intended as a conceptual map for the present discussion, not as a list of all capabilities implemented in RMG. In Coulomb gauge, $\nabla \cdot \mathbf{A} = 0$, the vector potential is transverse; after the longitudinal/transverse decomposition of the electric field, $-\nabla\Phi$ gives the longitudinal electric field, while $-(1/c)\partial\mathbf{A}/\partial t$ gives the transverse electric field and $\nabla \times \mathbf{A}$ gives the magnetic field. The present work addresses the homogeneous periodic optical-response row of Table 1. The finite- $\mathbf{q}$, magnetic, and multipolar rows are included only as broader perspective for related TDDFT response problems.^{10,12}

RT-TDDFT for extended systems has been developed in a variety of numerical representations. Real-space implementations such as Octopus provide time propagation for

finite and extended systems and serve here as the independent reference for cross-code validation.¹³ Plane-wave and related periodic approaches, including `Qbox/Qball` developments and `CP2K`, have demonstrated real-time response calculations under periodic boundary conditions.^{10,14–16} All-electron and atom-centered formulations provide complementary routes for crystalline and molecular systems, including recent periodic RT-TDDFT implementations in `exciting`, numeric atom-centered basis frameworks, and `PySCF`.^{11,17,18} Finite-element approaches provide another real-space route to large systems.¹⁹ In a related approximate direction, Wong and co-workers have developed a velocity-gauge real-time TD-DFTB implementation for periodic condensed-matter systems based on the DFTB+ framework.^{20,21} These developments show that periodic real-time response is an active area with multiple complementary implementations; the present work adds the particular combination of periodic current response, collinear spin resolution, finite active-space density-matrix propagation, and real-space multigrid scalability in `RMG`.

Building on our previous finite-system `RMG` RT-TDDFT implementation, which demonstrated real-time simulations of systems containing up to approximately 24,000 electrons on massively parallel architectures,²² the present work extends the framework to periodic boundary conditions and collinear spin-polarized response. This extension is not only a change of boundary conditions. Periodic materials require Bloch-orbital and k-point resolution, a current-based optical observable, and careful treatment of nonlocal pseudopotential contributions to the current operator. Spin-polarized systems further require separate density matrices and self-consistent densities for the two collinear spin channels. The implementation also builds on the real-space multigrid infrastructure of `RMG`, which is designed for large-scale parallel calculations on modern CPU and GPU architectures.²³

The main contributions of this paper are as follows. We formulate periodic RT-TDDFT for homogeneous optical response in the velocity gauge, with the macroscopic response expressed through the current. We derive the periodic current operator, including the contribution from norm-conserving nonlocal pseudopotentials. We extend the density-matrix propagation scheme to finite active spaces resolved by Bloch wave vector and collinear spin channel. We then describe the corresponding implementation in `RMG`, including k-point, spin-channel, real-space, and distributed-matrix parallel organization.

We validate the periodic and spin-polarized implementation by comparing current-derived optical spectra from `RMG` with independent RT-TDDFT calculations for a sequence of increasingly demanding systems: a one-dimensional periodic hydrogen-dimer reference, pristine monolayer hBN, bulk Si, and spin-polarized CrI_3 . These comparisons test the principal spectral structure rather than absolute optical intensity. As a large-scale scientific demonstration, we apply the method to the negatively charged nitrogen-vacancy center in diamond, a spin-polarized defect whose optical response probes local defect orbitals, polarization selection rules, spin-channel selectivity, active-space convergence, and defect-concentration effects.^{24,25} The largest calculation contains a single NV^- center in a 4096-site diamond framework, corresponding to 4095 atoms after defect formation, and approaches the dilute-defect limit.

The remainder of the manuscript is organized as follows. Section 2 develops the periodic velocity-gauge formalism, the current operator including nonlocal pseudopotential contributions, density-matrix propagation, the collinear spin extension, and the formal current-based optical response. Section 3 describes the current numerical realization in `RMG`, including

implementation-specific choices in the perturbation, current evaluation, propagation, parallel organization, and spectral processing. Section 4 presents the H₂, hBN, Si, and CrI₃ validation benchmarks. Section 5 presents the NV⁻-center application, including selection-rule tests, active-space dependence, concentration effects, the 4095-atom single-defect calculation, and collinear magnetic configuration comparisons.

2 Theory and Methodology

2.1 Periodic RT-TDDFT and the Velocity Gauge

The starting point for the present periodic RT-TDDFT formulation is the time-dependent Kohn–Sham equation in the presence of an external electromagnetic perturbation,

$$i\hbar \frac{\partial}{\partial t} \Psi_j(\mathbf{r}, t) = \hat{H}(t) \Psi_j(\mathbf{r}, t), \quad (1)$$

where

$$\hat{H}(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + q_e \Phi(\mathbf{r}, t) + \hat{V}_{\text{KS}}[\rho](t). \quad (2)$$

Here $q_e = -e$ is the electron charge, with $e > 0$, m_e is the electron mass, c is the speed of light, and

$$\hat{\mathbf{p}} = -i\hbar \nabla \quad (3)$$

is the canonical momentum operator. Thus, for an electron, the vector potential enters the kinetic momentum as $\hat{\mathbf{p}} + e\mathbf{A}/c$. In the norm-conserving pseudopotential formulation used here, the Kohn–Sham potential is decomposed as

$$\hat{V}_{\text{KS}}[\rho](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{NL}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}[\rho](t), \quad (4)$$

where $\hat{V}_{\text{ion}}^{\text{loc}}$ is the local ionic pseudopotential, $\hat{V}_{\text{NL}}$ is the nonlocal pseudopotential operator, $\hat{V}_{\text{H}}$ is the Hartree potential, and $\hat{V}_{\text{xc}}$ is the exchange-correlation potential.

The physical electric and magnetic fields are obtained from the electromagnetic potentials according to

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \Phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (5)$$

Different choices of Φ and $\mathbf{A}$ related by a gauge transformation describe the same electromagnetic fields. This gauge freedom should be distinguished from the long-wavelength approximation used for optical response. In the present work we consider the homogeneous-field limit, in which the spatial variation of the external field over the simulation cell is neglected. This approximation corresponds to the optical, $\mathbf{q} \rightarrow 0$, limit and leads to vertical transitions at fixed crystal momentum. The $\mathbf{q} \rightarrow 0$ restriction therefore arises from the homogeneous-field approximation, not from the choice of velocity gauge.

For finite systems, a spatially uniform electric field is commonly represented in the length gauge by setting

$$\mathbf{A}_L(\mathbf{r}, t) = 0, \quad \Phi_L(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{E}(t). \quad (6)$$

The corresponding scalar-potential energy for an electron is $q_e \Phi_L(\mathbf{r}, t) = e \mathbf{r} \cdot \mathbf{E}(t)$. This representation is well suited to finite molecular systems, where the time-dependent dipole moment is a well-defined observable. Under periodic boundary conditions, however, the position operator is not compatible with Born–von Karman periodicity: $\mathbf{r}$ and $\mathbf{r} + \mathbf{L}$ represent equivalent points of the crystal but give different values of the scalar potential in Eq. (6). The length-gauge scalar potential is therefore not a periodic operator.

The same homogeneous electric field can be represented in the velocity gauge by a gauge transformation. For a scalar gauge function $\chi(\mathbf{r}, t)$,

$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \nabla \chi(\mathbf{r}, t), \quad \Phi'(\mathbf{r}, t) = \Phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \chi(\mathbf{r}, t)}{\partial t}, \quad (7)$$

and the electronic wavefunction transforms as

$$\Psi'(\mathbf{r}, t) = \exp \left[\frac{iq_e}{\hbar c} \chi(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t). \quad (8)$$

Starting from the length-gauge potentials in Eq. (6), choose

$$\chi(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{A}(t), \quad \frac{d\mathbf{A}(t)}{dt} = -c\mathbf{E}(t). \quad (9)$$

This gives

$$\Phi_V(\mathbf{r}, t) = 0, \quad \mathbf{A}_V(\mathbf{r}, t) = \mathbf{A}(t), \quad \mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}. \quad (10)$$

For an electron, the corresponding unitary transformation is

$$\hat{U}(t) = \exp \left[-\frac{ie}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right], \quad \Psi_V(\mathbf{r}, t) = \hat{U}(t) \Psi_L(\mathbf{r}, t). \quad (11)$$

Because $\mathbf{A}(t)$ is spatially uniform, $\nabla \times \mathbf{A}(t) = 0$; the homogeneous velocity-gauge field is therefore the gauge-equivalent representation of the electric-dipole field, not a full finite- $\mathbf{q}$ transverse electromagnetic wave. Finite- $\mathbf{q}$ response functions lie outside the homogeneous-field formulation considered here.

This relation also fixes the formal connection between a conventional broadband electric-field kick and the corresponding velocity-gauge vector potential. For an impulsive perturbation,

$$\mathbf{E}(t) = \mathbf{E}_0 \delta(t), \quad (12)$$

integration of Eq. (10) gives, for $\mathbf{A}(t < 0) = 0$ under the sign conventions established above,

$$\mathbf{A}(t) = -c\mathbf{E}_0 \Theta(t). \quad (13)$$

The particular impulsive realization used in RMG is described separately in Sec. 3.

In the formal velocity-gauge representation, the periodic TDKS Hamiltonian for an electron is

$$\hat{H}_V(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL}}^V(t). \quad (14)$$

Here

$$\hat{V}_{\text{loc}}[\rho](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}[\rho](t) \quad (15)$$

contains all local terms, while the nonlocal pseudopotential is written separately. Since $q_e = -e$, the kinetic momentum in Eq. (14) is $\hat{\mathbf{p}} + e\mathbf{A}(t)/c$ for an electron. If a nonlocal pseudopotential is present, it must be transformed consistently with the same unitary gauge transformation,

$$\hat{V}_{\text{NL}}^V(t) = \hat{U}(t)\hat{V}_{\text{NL}}\hat{U}^\dagger(t), \quad \hat{U}(t) = \exp\left[\frac{iq_e}{\hbar c}\mathbf{r} \cdot \mathbf{A}(t)\right]. \quad (16)$$

The detailed consequences of Eq. (16) for the current operator are discussed in Sec. 2.2.

For a periodic system, the time-dependent Kohn–Sham orbitals are written in Bloch form,

$$\Psi_{n\mathbf{k}}(\mathbf{r}, t) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}, t), \quad (17)$$

where $u_{n\mathbf{k}}(\mathbf{r}, t)$ has the periodicity of the unit cell. Acting on a Bloch orbital, the canonical momentum becomes

$$\hat{\mathbf{p}}\Psi_{n\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{r}} (-i\hbar\nabla + \hbar\mathbf{k}) u_{n\mathbf{k}}. \quad (18)$$

The corresponding cell-periodic equation is

$$i\hbar \frac{\partial}{\partial t} u_{n\mathbf{k}}(\mathbf{r}, t) = \hat{H}_{\mathbf{k}}^V(t) u_{n\mathbf{k}}(\mathbf{r}, t), \quad (19)$$

with

$$\hat{H}_{\mathbf{k}}^V(t) = \frac{1}{2m_e} \left[-i\hbar\nabla + \hbar\mathbf{k} - \frac{q_e}{c}\mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL},\mathbf{k}}^V(t). \quad (20)$$

Equivalently, using $q_e = -e$, the kinetic operator contains $-i\hbar\nabla + \hbar\mathbf{k} + e\mathbf{A}(t)/c$. Equation (20) is the periodic counterpart of the finite-system length-gauge RT-TDDFT formulation used in our previous work,²² but with the position-dependent dipole perturbation replaced by a periodic velocity-gauge coupling.

Because the dipole moment is not a well-defined bulk observable under periodic boundary conditions, the natural time-dependent observable for periodic optical response is the macroscopic current rather than the molecular dipole moment. The construction of the current operator, including the nonlocal pseudopotential contribution, is given in Sec. 2.2; the extraction of optical response functions from the real-time current is described in Sec. 2.5.

2.2 Current Operator and Nonlocal Pseudopotentials

For a periodic system in the homogeneous-field limit, the optical response is naturally formulated in terms of the macroscopic charge-current density. We define

$$J_\alpha(t) = \frac{1}{V_{\text{cell}}} \sum_{\mathbf{k}, \sigma} w_{\mathbf{k}} \text{Tr} \left[\mathbf{P}_{\mathbf{k}}^\sigma(t) \hat{J}_{\mathbf{k}, \alpha}^\sigma(t) \right], \quad (21)$$

where V_{cell} is the volume of the periodic simulation cell, which may be a primitive cell or a supercell, $w_{\mathbf{k}}$ are the normalized Brillouin-zone weights satisfying $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$, $\mathbf{P}_{\mathbf{k}}^\sigma$ is the

single-particle density matrix for spin channel σ , and

$$\hat{I}_\alpha(t) = q_e \hat{v}_\alpha(t) \quad (22)$$

is the charge-current operator. The trace in Eq. (21) is taken over the propagated single-particle subspace at fixed $\mathbf{k}$ and spin.

The velocity operator is most cleanly defined before introducing the cell-periodic Bloch representation. For the velocity-gauge Hamiltonian of Eq. (14),

$$\hat{v}_\alpha(t) = \frac{i}{\hbar} \left[\hat{H}_V(t), \hat{r}_\alpha \right]. \quad (23)$$

Since the local potential commutes with the position operator, the Hamiltonian containing a local potential and a nonlocal pseudopotential gives

$$\hat{v}_\alpha(t) = \frac{1}{m_e} \left[\hat{p}_\alpha - \frac{q_e}{c} A_\alpha(t) \right] + \frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_\alpha \right]. \quad (24)$$

Equivalently,

$$\frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_\alpha \right] = \frac{1}{i\hbar} \left[\hat{r}_\alpha, \hat{V}_{\text{NL}}^V(t) \right], \quad (25)$$

which fixes the sign convention for the nonlocal contribution. Thus the formal charge-current operator is

$$\hat{I}_\alpha(t) = q_e \left\{ \frac{1}{m_e} \left[\hat{p}_\alpha - \frac{q_e}{c} A_\alpha(t) \right] + \frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_\alpha \right] \right\}. \quad (26)$$

The explicit $A_\alpha(t)$ dependence of the kinetic term gives the usual diamagnetic contribution,

$$\hat{I}_\alpha^{\text{dia}}(t) = -\frac{q_e^2}{m_e c} A_\alpha(t) \hat{1}, \quad (27)$$

while the nonlocal term is also A dependent through the gauge-transformed nonlocal pseudopotential.

The same nonlocal commutator appears when the velocity-gauge Hamiltonian is expanded to first order in the homogeneous vector potential. Using Eq. (16),

$$\hat{V}_{\text{NL}}^V(t) = \hat{U}(t) \hat{V}_{\text{NL}} \hat{U}^\dagger(t), \quad \hat{U}(t) = \exp \left[\frac{iq_e}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right], \quad (28)$$

one obtains

$$\hat{V}_{\text{NL}}^V(t) = \hat{V}_{\text{NL}} + \frac{iq_e}{\hbar c} \sum_\alpha A_\alpha(t) \left[\hat{r}_\alpha, \hat{V}_{\text{NL}} \right] + \mathcal{O}(A^2). \quad (29)$$

The first-order nonlocal perturbation is therefore

$$\delta \hat{V}_{\text{NL}}^{(1)}(t) = -\frac{1}{c} \sum_\alpha A_\alpha(t) \hat{I}_{\alpha, \text{NL}}^{(0)}, \quad \hat{I}_{\alpha, \text{NL}}^{(0)} = \frac{q_e}{i\hbar} \left[\hat{r}_\alpha, \hat{V}_{\text{NL}} \right]. \quad (30)$$

Combining this term with the kinetic part of the minimal-coupling Hamiltonian gives the

linear velocity-gauge perturbation

$$\delta\hat{H}^{(1)}(t) = -\frac{1}{c} \sum_{\alpha} A_{\alpha}(t) \hat{I}_{\alpha}^{(0)}, \quad (31)$$

where

$$\hat{I}_{\alpha}^{(0)} = q_e \left[\frac{\hat{p}_{\alpha}}{m_e} + \frac{i}{\hbar} [\hat{V}_{\text{NL}}, \hat{r}_{\alpha}] \right] = \frac{q_e}{m_e} \hat{p}_{\alpha} + \frac{q_e}{i\hbar} [\hat{r}_{\alpha}, \hat{V}_{\text{NL}}]. \quad (32)$$

Equations (26)–(32) show that the nonlocal-pseudopotential commutator is not only a correction to the measured current; it is also the nonlocal part of the linear velocity-gauge perturbation required for gauge consistency when nonlocal pseudopotentials are present.¹⁰ More generally, a nonlocal one-particle operator present in the Hamiltonian contributes to the velocity through the corresponding commutator with the position operator.

For a separable norm-conserving pseudopotential,²⁶

$$\hat{V}_{\text{NL}} = \sum_{I,lm} |\beta_{I,lm}\rangle D_{I,lm} \langle\beta_{I,lm}|, \quad (33)$$

the coordinate-space matrix element of the commutator can be written as

$$\langle \mathbf{r} | [\hat{r}_{\alpha}, \hat{V}_{\text{NL}}] | \mathbf{r}' \rangle = (r_{\alpha} - r'_{\alpha}) V_{\text{NL}}(\mathbf{r}, \mathbf{r}'). \quad (34)$$

Equation (34) makes explicit that this term vanishes for a strictly local potential but is finite for a nonlocal pseudopotential.

The corresponding cell-periodic Bloch representation is obtained only after the full-space current operator has been defined. For $\Psi_{n\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}$, the canonical momentum becomes $-i\hbar\nabla + \hbar\mathbf{k}$ as in Eq. (18). The nonlocal operator at fixed $\mathbf{k}$ is

$$\hat{V}_{\text{NL},\mathbf{k}} = e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{V}_{\text{NL}} e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (35)$$

At zero external vector potential, the charge-current matrix in a cell-periodic basis is therefore

$$\begin{aligned} I_{ab,\mathbf{k},\alpha}^{(0),\sigma} &= \frac{q_e}{m_e} \langle u_{a\mathbf{k}}^{\sigma} | -i\hbar\nabla_{\alpha} + \hbar k_{\alpha} | u_{b\mathbf{k}}^{\sigma} \rangle_{\text{cell}} \\ &+ \frac{q_e}{i\hbar} \left\langle u_{a\mathbf{k}}^{\sigma} \left| \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL},\mathbf{k}}^{\sigma} \right] \right| u_{b\mathbf{k}}^{\sigma} \right\rangle_{\text{cell}}. \end{aligned} \quad (36)$$

The first term contains both the gradient and Bloch $\hbar k_{\alpha}$ contributions, while the second term is the nonlocal-pseudopotential contribution. In the presence of a finite homogeneous vector potential, Eq. (26) gives the corresponding A -dependent current operator, including the diamagnetic kinetic term and the gauge-transformed nonlocal operator.

Equations (21) and (36) define the formal current observable used in the periodic RT-TDDFT theory. The implementation-specific construction of the current matrices, the particular impulsive realization of the velocity-gauge perturbation, and the conversion of raw current traces to optical response functions are described separately in Secs. 3 and 2.5.

2.3 Density-Matrix Time Propagation

The time propagation used in this work builds on the Liouville-von Neumann density-matrix propagation framework introduced in earlier work²⁷ and on the finite-system RMG RT-TDDFT implementation developed in our previous work,²² with the extension that the propagated matrices are now resolved by crystal momentum and collinear spin channel. For each $\mathbf{k}$ and spin σ , the propagation is carried out in a finite subspace $\mathcal{A}_{\mathbf{k}}^{\sigma}$ spanned by ground-state Kohn–Sham Bloch orbitals,

$$\Psi_{a\mathbf{k}}^{\sigma}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{a\mathbf{k}}^{\sigma}(\mathbf{r}), \quad a \in \mathcal{A}_{\mathbf{k}}^{\sigma}. \quad (37)$$

The one-particle density matrix in this basis is

$$P_{ab,\mathbf{k}}^{\sigma}(t) = \langle u_{a\mathbf{k}}^{\sigma} | \hat{\rho}^{\sigma}(t) | u_{b\mathbf{k}}^{\sigma} \rangle_{\text{cell}}. \quad (38)$$

The initial condition is the ground-state occupation matrix,

$$P_{ab,\mathbf{k}}^{\sigma}(0) = f_{a\mathbf{k}}^{\sigma} \delta_{ab}, \quad (39)$$

where $f_{a\mathbf{k}}^{\sigma}$ is the ground-state occupation of state a at $\mathbf{k}$ in spin channel σ .

The density matrix obeys the Liouville-von Neumann equation

$$i\hbar \frac{d\mathbf{P}_{\mathbf{k}}^{\sigma}(t)}{dt} = [\mathbf{H}_{\mathbf{k}}^{\sigma}(t), \mathbf{P}_{\mathbf{k}}^{\sigma}(t)], \quad (40)$$

where

$$H_{ab,\mathbf{k}}^{\sigma}(t) = \left\langle u_{a\mathbf{k}}^{\sigma} \left| \hat{H}_{\mathbf{k}}^{V,\sigma}(t) \right| u_{b\mathbf{k}}^{\sigma} \right\rangle_{\text{cell}} \quad (41)$$

is the matrix representation of the velocity-gauge Kohn–Sham Hamiltonian of Eq. (20). At nonzero $\mathbf{k}$, both $\mathbf{H}_{\mathbf{k}}^{\sigma}$ and $\mathbf{P}_{\mathbf{k}}^{\sigma}$ are generally complex Hermitian matrices.

Formally, Eq. (40) is solved by a unitary time-evolution operator,

$$\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) = \mathcal{U}_{\mathbf{k}}^{\sigma}(t + \Delta t, t) \mathbf{P}_{\mathbf{k}}^{\sigma}(t) \mathcal{U}_{\mathbf{k}}^{\sigma\dagger}(t + \Delta t, t), \quad (42)$$

with

$$\mathcal{U}_{\mathbf{k}}^{\sigma}(t + \Delta t, t) = \exp[\Omega_{\mathbf{k}}^{\sigma}(t + \Delta t, t)]. \quad (43)$$

The Magnus generator²⁸ is written as

$$\Omega_{\mathbf{k}}^{\sigma} = \Omega_{1,\mathbf{k}}^{\sigma} + \Omega_{2,\mathbf{k}}^{\sigma} + \cdots, \quad (44)$$

where

$$\Omega_{1,\mathbf{k}}^{\sigma} = -\frac{i}{\hbar} \int_t^{t+\Delta t} \mathbf{H}_{\mathbf{k}}^{\sigma}(\tau) d\tau, \quad (45)$$

and

$$\Omega_{2,\mathbf{k}}^{\sigma} = -\frac{1}{2\hbar^2} \int_t^{t+\Delta t} d\tau_1 \int_t^{\tau_1} d\tau_2 [\mathbf{H}_{\mathbf{k}}^{\sigma}(\tau_1), \mathbf{H}_{\mathbf{k}}^{\sigma}(\tau_2)]. \quad (46)$$

In the present method, the Magnus expansion is truncated after its first term, and the time

integral of the Hamiltonian is evaluated using an endpoint-averaged, midpoint-equivalent approximation,

$$\Omega_{\mathbf{k}}^{\sigma}(t + \Delta t, t) \simeq -\frac{i\Delta t}{2\hbar} [\mathbf{H}_{\mathbf{k}}^{\sigma}(t) + \mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)]. \quad (47)$$

This approximation has the time symmetry associated with midpoint or trapezoidal integration of the first Magnus term. The detailed construction of $\mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)$ in the self-consistent propagation loop is described in Sec. 3.

Because $\mathbf{H}_{\mathbf{k}}^{\sigma}$ is Hermitian, the generator $\Omega_{\mathbf{k}}^{\sigma}$ is anti-Hermitian and $\mathcal{U}_{\mathbf{k}}^{\sigma}$ is unitary. The action of the unitary transformation on the density matrix can therefore be written as a Baker–Campbell–Hausdorff nested-commutator series,

$$\begin{aligned} \mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) &= e^{\Omega_{\mathbf{k}}^{\sigma}} \mathbf{P}_{\mathbf{k}}^{\sigma}(t) e^{-\Omega_{\mathbf{k}}^{\sigma}} \\ &= \mathbf{P}_{\mathbf{k}}^{\sigma}(t) + [\Omega_{\mathbf{k}}^{\sigma}, \mathbf{P}_{\mathbf{k}}^{\sigma}(t)] + \frac{1}{2!} [\Omega_{\mathbf{k}}^{\sigma}, [\Omega_{\mathbf{k}}^{\sigma}, \mathbf{P}_{\mathbf{k}}^{\sigma}(t)]] + \cdots \end{aligned} \quad (48)$$

Equivalently, defining $\mathbf{C}_{0,\mathbf{k}}^{\sigma} = \mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ and

$$\mathbf{C}_{n,\mathbf{k}}^{\sigma} = \frac{1}{n} [\Omega_{\mathbf{k}}^{\sigma}, \mathbf{C}_{n-1,\mathbf{k}}^{\sigma}], \quad (49)$$

one obtains

$$\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) = \sum_{n=0}^{\infty} \mathbf{C}_{n,\mathbf{k}}^{\sigma}. \quad (50)$$

This form applies equally to real Γ -point matrices and to the complex Hermitian matrices that arise for general $\mathbf{k}$ points. Exact unitary propagation conserves the trace, Hermiticity, and eigenvalues of each density matrix; therefore the single-particle occupations are preserved by the continuous evolution. In practical calculations these properties are preserved up to the numerical truncation and self-consistency errors of the propagation.

The propagated density matrices determine the real-space spin densities. When the propagated subspace is supplemented by frozen occupied states, the real-space spin density can be written as

$$\begin{aligned} \rho^{\sigma}(\mathbf{r}, t) &= \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{a,b \in \mathcal{A}_{\mathbf{k}}^{\sigma}} u_{a\mathbf{k}}^{\sigma}(\mathbf{r}) P_{ab,\mathbf{k}}^{\sigma}(t) u_{b\mathbf{k}}^{\sigma*}(\mathbf{r}) \\ &\quad + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{f \in \mathcal{F}_{\mathbf{k}}^{\sigma}} f_{f\mathbf{k}}^{\sigma} |u_{f\mathbf{k}}^{\sigma}(\mathbf{r})|^2. \end{aligned} \quad (51)$$

Here $\mathcal{F}_{\mathbf{k}}^{\sigma}$ denotes states that are not included in the propagated active subspace and remain fixed at their ground-state occupations. Equivalently, the same density can be written as a ground-state density plus the active-space density change,

$$\rho^{\sigma}(\mathbf{r}, t) = \rho_{\text{gs}}^{\sigma}(\mathbf{r}) + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{a,b \in \mathcal{A}_{\mathbf{k}}^{\sigma}} u_{a\mathbf{k}}^{\sigma}(\mathbf{r}) \Delta P_{ab,\mathbf{k}}^{\sigma}(t) u_{b\mathbf{k}}^{\sigma*}(\mathbf{r}), \quad (52)$$

where $\Delta \mathbf{P}_{\mathbf{k}}^{\sigma}(t) = \mathbf{P}_{\mathbf{k}}^{\sigma}(t) - \mathbf{P}_{\mathbf{k}}^{\sigma}(0)$. This form is especially useful for connecting the formal

density reconstruction to an active-space implementation with frozen states.

The propagation is self-consistent because the Hamiltonian is a functional of the evolving density. The update cycle can be summarized as

$$\{\mathbf{P}_{\mathbf{k}}^{\sigma}(t)\} \rightarrow \{\rho^{\sigma}(\mathbf{r}, t)\} \rightarrow \hat{V}_{\text{H}}[\rho](t), \hat{V}_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}](t) \rightarrow \{\mathbf{H}_{\mathbf{k}}^{\sigma}(t)\} \rightarrow \{\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t)\}. \quad (53)$$

The two collinear spin channels are propagated as separate density matrices, but they are not independent: both contribute to the total density entering the Hartree potential, and the exchange-correlation potential in each spin channel depends on $\rho^{\uparrow}$ and $\rho^{\downarrow}$. The spin structure is discussed in more detail in Sec. 2.4.

2.4 Collinear Spin Extension

The spin-polarized extension used in the present work is collinear. A fixed spin quantization axis is chosen, and the spin index $\sigma \in \{\uparrow, \downarrow\}$ labels two spin channels. For each $\mathbf{k}$ point, the one-particle density matrix and Hamiltonian have a block-diagonal spin structure,

$$\mathbf{P}_{\mathbf{k}}(t) = \begin{pmatrix} \mathbf{P}_{\mathbf{k}}^{\uparrow}(t) & 0 \\ 0 & \mathbf{P}_{\mathbf{k}}^{\downarrow}(t) \end{pmatrix}, \quad \mathbf{H}_{\mathbf{k}}(t) = \begin{pmatrix} \mathbf{H}_{\mathbf{k}}^{\uparrow}(t) & 0 \\ 0 & \mathbf{H}_{\mathbf{k}}^{\downarrow}(t) \end{pmatrix}. \quad (54)$$

Thus there are no off-diagonal spinor blocks in the present formulation. The two spin channels are propagated as separate density matrices, each obeying the Liouville-von Neumann equation introduced in Sec. 2.3,

$$i\hbar \frac{d\mathbf{P}_{\mathbf{k}}^{\sigma}(t)}{dt} = [\mathbf{H}_{\mathbf{k}}^{\sigma}(t), \mathbf{P}_{\mathbf{k}}^{\sigma}(t)]. \quad (55)$$

The spin-resolved electron densities are reconstructed from $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ as described in Sec. 2.3. The total electron density is

$$\rho(\mathbf{r}, t) = \rho^{\uparrow}(\mathbf{r}, t) + \rho^{\downarrow}(\mathbf{r}, t). \quad (56)$$

In the collinear Kohn–Sham Hamiltonian, the local effective potential for spin channel σ can be written schematically as

$$\hat{V}_{\text{loc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}](t). \quad (57)$$

The Hartree potential depends on the total electron density $\rho = \rho^{\uparrow} + \rho^{\downarrow}$, while the exchange-correlation potential is spin-channel dependent and is a functional of both spin densities. In the absence of spin-orbit coupling, the nonlocal pseudopotential terms considered here are diagonal in the collinear spin basis and are treated as described in Secs. 2.1 and 2.2.

Equation (55) therefore represents separate propagation within each spin channel, but the channels are not physically independent. They are coupled self-consistently through the density-dependent potentials:

$$\{\mathbf{P}_{\mathbf{k}}^{\uparrow}, \mathbf{P}_{\mathbf{k}}^{\downarrow}\} \rightarrow \{\rho^{\uparrow}, \rho^{\downarrow}\} \rightarrow \hat{V}_{\text{H}}[\rho^{\uparrow} + \rho^{\downarrow}], \hat{V}_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}] \rightarrow \{\mathbf{H}_{\mathbf{k}}^{\uparrow}, \mathbf{H}_{\mathbf{k}}^{\downarrow}\}. \quad (58)$$

The same spin decomposition applies to the macroscopic charge current. For each spin channel,

$$J_{\alpha}^{\sigma}(t) = \frac{1}{V_{\text{cell}}} \sum_{\mathbf{k}} w_{\mathbf{k}} \text{Tr} \left[\mathbf{P}_{\mathbf{k}}^{\sigma}(t) \hat{I}_{\mathbf{k},\alpha}^{\sigma}(t) \right], \quad (59)$$

where the charge-current operator is the spin-diagonal operator defined in Sec. 2.2. The total macroscopic charge-current density is the sum over the two collinear spin channels,

$$J_{\alpha}(t) = \sum_{\sigma=\uparrow,\downarrow} J_{\alpha}^{\sigma}(t). \quad (60)$$

The individual $J_{\alpha}^{\sigma}(t)$ may also be analyzed separately to obtain spin-resolved optical response within the fixed collinear basis.

The present formulation does not include off-diagonal spin coherences, transverse spin dynamics, general two-component spinors, noncollinear magnetization dynamics, or spin-orbit-induced spin mixing. Such extensions require a spinor Hamiltonian with off-diagonal spin blocks and lie outside the collinear spin-polarized RT-TDDFT framework used in this manuscript. This spin-resolved collinear formulation provides the theoretical basis for analyzing the spin-up and spin-down optical channels of the NV⁻ center discussed below.

2.5 Optical Response and Postprocessing

The velocity-gauge formulation developed above is not restricted to an impulsive perturbation. For a prescribed spatially homogeneous vector potential $\mathbf{A}(t)$, the formal Hamiltonian retains the external field for all times at which the chosen electromagnetic perturbation requires it,

$$\hat{H}_V(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL}}^V(t). \quad (61)$$

Thus the same formal propagation can describe an impulsive excitation, a finite-duration pulse, or a persistent time-dependent driving field, provided the corresponding $\mathbf{A}(t)$ is used in the Hamiltonian and in the current operator of Sec. 2.2.

We use the Fourier-transform convention

$$f(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\omega) e^{i\omega t} d\omega. \quad (62)$$

With this convention,

$$\mathcal{F} \left[\frac{df}{dt} \right] = i\omega f(\omega), \quad (63)$$

assuming that boundary terms vanish or are treated separately. In the homogeneous velocity gauge,

$$\mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}, \quad \mathbf{A}(t) = \mathbf{A}(t_0) - c \int_{t_0}^t \mathbf{E}(t') dt', \quad (64)$$

and therefore

$$E_{\alpha}(\omega) = -\frac{i\omega}{c} A_{\alpha}(\omega). \quad (65)$$

A conventional broadband electric-field kick used in real-time linear response calculations^{3,22} is

$$E_\beta(t) = E_{0,\beta}\delta(t). \quad (66)$$

For $A_\beta(t < 0) = 0$, Eq. (64) gives

$$A_\beta(t) = -cE_{0,\beta}\Theta(t). \quad (67)$$

Thus, in the formally equivalent velocity-gauge representation, the vector potential remains at its post-kick value after the instantaneous electric-field impulse.

The time-dependent observable is the physical macroscopic charge-current density defined in Sec. 2.2. We write the induced current as

$$\Delta J_\alpha(t) = J_\alpha(t) - J_\alpha^{\text{eq}}, \quad (68)$$

where J_α^{eq} is the equilibrium current before the external perturbation. For the non-current-carrying equilibrium states considered here, the macroscopic equilibrium current vanishes in the exact ground state. Retaining the induced-current notation is nevertheless useful in numerical calculations, where finite $\mathbf{k}$ -point sampling and numerical errors may produce a small residual baseline.

In the linear-response regime,

$$\Delta J_\alpha(\omega) = \sum_\beta \sigma_{\alpha\beta}(\omega) E_\beta(\omega), \quad (69)$$

which defines the optical conductivity tensor $\sigma_{\alpha\beta}(\omega)$. Separate perturbations along linearly independent Cartesian directions may be used to construct the full tensor. For the electric-field impulse of Eq. (66), $E_\beta(\omega) = E_{0,\beta}$, and therefore

$$\sigma_{\alpha\beta}(\omega) = \frac{\Delta J_\alpha^{(\beta)}(\omega)}{E_{0,\beta}}, \quad (70)$$

where $\Delta J_\alpha^{(\beta)}$ denotes the current response to a kick polarized along direction β .

The dielectric response follows from the relation between current and macroscopic polarization. Defining the induced polarization by

$$\Delta J_\alpha(t) = \frac{d\Delta P_\alpha(t)}{dt}, \quad (71)$$

one obtains

$$\Delta J_\alpha(\omega) = i\omega \Delta P_\alpha(\omega), \quad \Delta P_\alpha(\omega) = \frac{\Delta J_\alpha(\omega)}{i\omega}. \quad (72)$$

Thus integration of the current in time corresponds formally to division by $i\omega$ in frequency space, with the zero-frequency component requiring separate treatment.

Using Gaussian electromagnetic units, the displacement field satisfies

$$D_\alpha(\omega) = E_\alpha(\omega) + 4\pi P_\alpha(\omega) = \sum_\beta \varepsilon_{\alpha\beta}(\omega) E_\beta(\omega). \quad (73)$$

Combining Eqs. (69) and (72) gives

$$\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} - \frac{4\pi\iota}{\omega} \sigma_{\alpha\beta}(\omega). \quad (74)$$

With the Fourier convention of Eq. (62), the dissipative diagonal response is represented by $\text{Re } \sigma_{\alpha\alpha}(\omega)$ or, equivalently, by $-\text{Im } \varepsilon_{\alpha\alpha}(\omega)$. The sign of the imaginary part follows from the $e^{-\iota\omega t}$ transform convention used here.

The conversion from dielectric response to an absorption coefficient or absorbance depends on dimensionality and optical geometry. For a homogeneous bulk three-dimensional medium, one may introduce a complex refractive index $N_\alpha(\omega) = n_\alpha(\omega) + \iota\kappa_\alpha(\omega)$ along a principal axis, with $N_\alpha^2 = \varepsilon_{\alpha\alpha}$ and the sign of κ_α chosen to describe attenuation of the physical field. The corresponding bulk absorption coefficient is

$$\alpha_{\text{abs},\alpha}^{3\text{D}}(\omega) = \frac{2\omega}{c} \kappa_\alpha(\omega). \quad (75)$$

Absorbance additionally depends on the optical path length and sample geometry. Low-dimensional periodic systems, such as monolayers treated with supercell boundary conditions, require the appropriate geometry-dependent conversion. When only relative spectral features are compared, normalized absorption-like spectra should therefore be distinguished from quantitatively normalized conductivities, dielectric functions, absorption coefficients, or absorbances.

In a finite real-time calculation, the Fourier transform is evaluated over a finite propagation interval and is commonly combined with a window or damping function to control ringing and define an effective spectral broadening. A generic finite-time estimate of the induced current spectrum is

$$\Delta J_\alpha^T(\omega) = \int_0^T W(t) \Delta J_\alpha(t) e^{-\iota\omega t} dt, \quad (76)$$

where $W(t)$ is the chosen damping or window function. The finite propagation time and the choice of $W(t)$ determine the practical frequency resolution and broadening of the computed spectrum.

For collinear spin-polarized calculations, the total charge current is the sum of the spin-channel currents introduced in Sec. 2.4,

$$\Delta J_\alpha(t) = \sum_{\sigma=\uparrow,\downarrow} \Delta J_\alpha^\sigma(t). \quad (77)$$

The total optical conductivity follows from the total charge current in Eq. (69). Spin-channel-resolved spectra may also be formed from the separate $\Delta J_\alpha^\sigma(t)$ responses within the fixed

collinear spin basis; this does not imply spin-flip transitions or noncollinear spin dynamics.

Equations (64)–(76) define the formal current-based optical response used in the present periodic velocity-gauge RT-TDDFT theory.¹⁰ The numerical realization of the impulsive perturbation, the normalization of raw current traces, and the finite-time processing used in RMG are described in Sec. 3.

3 Implementation in RMG

The formal theory in Sec. 2.1–2.5 defines the periodic velocity-gauge RT-TDDFT formulation and the corresponding physical response functions. This section describes the current numerical realization in RMG. The implementation inherits the density-matrix propagation strategy of the previous finite-system RT-TDDFT module,²² and extends it to periodic boundary conditions, k-point sampling, collinear spin polarization, and current-based response analysis.

3.1 Numerical Implementation and Density-Matrix Propagation

A periodic RT-TDDFT calculation begins from a converged ground-state Kohn–Sham calculation. For each sampled $\mathbf{k}$ point and collinear spin channel σ , the ground-state cell-periodic Bloch orbitals are retained as a fixed basis for the real-time propagation. In this basis, RMG constructs and propagates finite active-space Hamiltonian and density matrices, $\mathbf{H}_{\mathbf{k}}^{\sigma}(t)$ and $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$, as introduced formally in Sec. 2.3. General $\mathbf{k}$ points lead to complex Hermitian matrices, while Γ -point calculations may use real-valued paths where applicable.

The implementation equations are evaluated internally in atomic units, with $\hbar = m_e = e = 1$ and $q_e = -1$. The speed of light is not set to unity by this convention. The relation between the explicit electromagnetic quantities in Sec. 2.1 and the numerical coefficient used by the VECTOR_POT mode is therefore treated separately in Sec. 3.2. The present periodic RT-TDDFT path is restricted to norm-conserving pseudopotentials.

The propagated orbital subspace is selected from the ground-state Kohn–Sham states. Its lower boundary is specified by the input parameter `tddft_start_state`; states below this boundary are not included in the propagated density matrix. Let $\{n_a\}$ denote the ground-state band indices retained in the active subspace and let N_{prop} be the number of propagated states. At the beginning of the real-time calculation,

$$H_{ab,\mathbf{k}}^{\sigma}(0) = \epsilon_{n_a\mathbf{k}}^{\sigma} \delta_{ab}, \quad P_{ab,\mathbf{k}}^{\sigma}(0) = f_{n_a\mathbf{k}}^{\sigma} \delta_{ab}, \quad (78)$$

where $\epsilon_{n_a\mathbf{k}}^{\sigma}$ and $f_{n_a\mathbf{k}}^{\sigma}$ are the ground-state eigenvalue and occupation of state n_a . Occupied states below the active lower boundary remain frozen at their ground-state density contribution. Occupied and unoccupied states inside the active subspace are included in $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ and can participate in the real-time response.

For a time step Δt , RMG uses the first term of the Magnus expansion with an endpoint-averaged Hamiltonian. In atomic units, the mathematically equivalent generator is

$$\Omega_{\mathbf{k}}^{\sigma}(t + \Delta t, t) \simeq -\frac{i\Delta t}{2} [\mathbf{H}_{\mathbf{k}}^{\sigma}(t) + \mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)]. \quad (79)$$

The code stores the Hermitian Hamiltonian integral separately from the factor of $-i$ used in the commutator propagation, but the resulting update is equivalent to

$$\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) = e^{\Omega_{\mathbf{k}}^{\sigma}} \mathbf{P}_{\mathbf{k}}^{\sigma}(t) e^{-\Omega_{\mathbf{k}}^{\sigma}}. \quad (80)$$

This unitary transformation is applied through the BCH nested-commutator expansion. The complex periodic implementation uses this BCH propagation route.

The endpoint Hamiltonian $\mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)$ is obtained self-consistently. RMG first predicts the endpoint Hamiltonian from the preceding Hamiltonian history. With this trial endpoint, Eq. (79) is used to propagate the density matrix, the real-space spin density is reconstructed, and the Hartree and spin-dependent exchange–correlation potentials are updated. The matrix elements of the resulting change in the local effective potential are then added to the endpoint Hamiltonian, and the cycle is repeated until the endpoint Hamiltonian is self consistent:

$$\mathbf{H}(t + \Delta t) \rightarrow \Omega \rightarrow \mathbf{P}(t + \Delta t) \rightarrow \rho^{\sigma}(\mathbf{r}, t + \Delta t) \rightarrow V_{\text{H}}[\rho], V_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}] \rightarrow \mathbf{H}(t + \Delta t).$$

The two collinear spin channels are propagated as separate density matrices, but the potentials are coupled through the total density entering the Hartree term and through the spin dependence of the exchange–correlation functional.

The real-space density reconstruction follows the active-space form of Sec. 2.3. RMG forms the active density-matrix change

$$\Delta P_{ab,\mathbf{k}}^{\sigma}(t) = P_{ab,\mathbf{k}}^{\sigma}(t) - f_{n_{a\mathbf{k}}}^{\sigma} \delta_{ab}, \quad (81)$$

and reconstructs

$$\rho^{\sigma}(\mathbf{r}, t) = \rho_{\text{gs}}^{\sigma}(\mathbf{r}) + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{ab=1}^{N_{\text{prop}}} u_{n_{a\mathbf{k}}}^{\sigma}(\mathbf{r}) \Delta P_{ab,\mathbf{k}}^{\sigma}(t) u_{n_{b\mathbf{k}}}^{\sigma*}(\mathbf{r}). \quad (82)$$

The ordering of the orbital factors in Eq. (82) matches the density-matrix convention of Sec. 2.3: the ground-state density supplies the frozen occupied contribution and the reference occupation of the active subspace, while $\Delta \mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ supplies the time-dependent active-space density change. This construction makes convergence with respect to the number of included virtual states an explicit numerical parameter of the RT-TDDFT calculation.

Exact unitary propagation preserves the trace, Hermiticity, and eigenvalues of each density matrix. In practical calculations these properties are maintained up to errors from the Magnus truncation, the finite BCH expansion, and the convergence tolerance of the endpoint self-consistency cycle.

3.2 External Perturbation and Current Evaluation

In the formal velocity-gauge theory, a prescribed homogeneous vector potential $\mathbf{A}(t)$ remains in the Hamiltonian for all times required by the external field. The present RMG VECTOR_POT mode implements a more specific discrete momentum-coupled excitation used to initiate the real-time response. We denote by κ_{α} the Cartesian numerical coefficient obtained from the

`electric_field_tddft` input after the code's coordinate conversion. This symbol denotes the coefficient actually used by the implementation; it is not identified here uniquely with a physical electric-field amplitude, vector-potential amplitude, A_α/c , or eA_α/c .

Before applying the perturbation, RMG constructs the kinetic/Bloch momentum-like matrix

$$M_{ab,\mathbf{k},\alpha}^{\text{kin},\sigma} = \imath \langle (\nabla_\alpha + \imath k_\alpha) u_{n_a\mathbf{k}}^\sigma | u_{n_b\mathbf{k}}^\sigma \rangle_{\text{cell}}. \quad (83)$$

Using integration by parts under periodic boundary conditions, this is equivalent to the cell-periodic matrix element of $-\imath\nabla_\alpha + k_\alpha$ in atomic units. The explicit $\mathbf{k}$ term is the Bloch contribution to the momentum operator.

The implemented initial perturbation is then

$$\Delta \mathbf{H}_{\mathbf{k}}^{\text{code},\sigma} = \sum_{\alpha} \kappa_{\alpha} \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma}. \quad (84)$$

The verified construction order is

$$\text{VecPHmatrix}() \rightarrow \text{VECTOR_POT perturbation} \rightarrow \text{CurrentNlpp}().$$

Consequently, the perturbing operator in Eq. (84) contains the kinetic/Bloch contribution only. The nonlocal-pseudopotential commutator correction does not enter the initial perturbing kick in the present implementation.

The perturbing contribution enters the first accepted midpoint propagator through the initial Hamiltonian endpoint and is absent from the converged self-consistent Hamiltonian without the external perturbing term at subsequent times. Since the first Magnus generator uses an endpoint-averaged Hamiltonian, the effective integrated first-order numerical action of the perturbation is

$$\int dt \Delta \mathbf{H}_{\mathbf{k}}^{\text{code},\sigma}(t) \approx \frac{\Delta t}{2} \sum_{\alpha} \kappa_{\alpha} \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma}. \quad (85)$$

Equation (85) describes the current discrete numerical realization. It should not be read as a statement that the code applies a mathematically exact Dirac-delta vector potential or the formally persistent step-like vector potential of Sec. 2.5.

After the perturbation is applied, RMG adds the nonlocal pseudopotential contribution to the matrices used for current evaluation. The routine `CurrentNlpp()` calls the same directional nonlocal-operator path used for the projector commutator. Expressed in the sign convention of Sec. 2.2, the atomic-unit nonlocal velocity contribution is $\imath[\hat{V}_{\text{NL}}, \hat{r}_\alpha]$, or equivalently $(1/\imath)[\hat{r}_\alpha, \hat{V}_{\text{NL}}]$. The corresponding raw matrix contribution is written as

$$M_{ab,\mathbf{k},\alpha}^{\text{NL},\sigma} = \frac{1}{\imath} \langle u_{n_a\mathbf{k}}^\sigma | [\hat{r}_\alpha, \hat{V}_{\text{NL},\mathbf{k}}^\sigma] | u_{n_b\mathbf{k}}^\sigma \rangle_{\text{cell}}. \quad (86)$$

Equivalently, this term may be written as $\imath \langle u_{n_a\mathbf{k}}^\sigma | [\hat{V}_{\text{NL},\mathbf{k}}^\sigma, \hat{r}_\alpha] | u_{n_b\mathbf{k}}^\sigma \rangle_{\text{cell}}$. This is the same sign convention used in the formal current matrix of Eq. (36), excluding the charge factor q_e and the volume normalization that are not included in the raw RMG matrix.

Thus the raw current-evaluation matrix used by RMG is

$$\mathbf{M}_{\mathbf{k},\alpha}^{\text{raw},\sigma} = \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma} + \mathbf{M}_{\mathbf{k},\alpha}^{\text{NL},\sigma}. \quad (87)$$

For calculations using the simple DFT+U implementation, the same directional operator-construction path also includes the corresponding Hubbard-projector contribution,

$$\mathbf{M}_{\mathbf{k},\alpha}^{\text{raw},\sigma} = \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma} + \mathbf{M}_{\mathbf{k},\alpha}^{\text{NL},\sigma} + \mathbf{M}_{\mathbf{k},\alpha}^{U,\sigma}. \quad (88)$$

This statement identifies the implemented code path; it does not by itself establish that the DFT+U contribution has been independently proven to be the full gauge-consistent commutator for a Hubbard functional.

The distinction between perturbation and measurement is therefore important. The current RMG `VECTOR_POT` perturbation uses the kinetic/Bloch momentum-like matrix in Eq. (84), whereas the matrix subsequently used to evaluate the current-like response contains the nonlocal pseudopotential correction and, where active, the simple DFT+U projector contribution. The implementation also does not explicitly construct the A -dependent diamagnetic current operator appearing in the formal charge-current operator of Sec. 2.2.

3.3 Periodic, Spin-Resolved, and Parallel Implementation

The periodic implementation propagates a separate active-space Hamiltonian and density matrix for each local $\mathbf{k}$ point and collinear spin channel. Thus $\mathbf{H}_{\mathbf{k}}^{\sigma}(t)$, $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$, and the corresponding momentum/current-like matrices are stored and updated independently for each $\mathbf{k}$ handled by a given MPI task group. General $\mathbf{k}$ points require complex Bloch orbitals and complex Hermitian matrices, while Γ -point calculations may use real-valued orbital and matrix paths where applicable. The Bloch $\mathbf{k}$ contribution to the kinetic momentum matrix is included in the operator construction described in Sec. 3.2.

The density and observable reductions use the Brillouin-zone weights associated with the sampled $\mathbf{k}$ points. During the self-consistent endpoint update, RMG reconstructs a density contribution from each local $\mathbf{k}$ point, multiplies it by the corresponding weight, and sums over the distributed $\mathbf{k}$ set. The same weighted reduction is used for current-like vector observables. After the $\mathbf{k}$ -point and matrix-distribution reductions, the vector current is passed through the RMG symmetry machinery for vector symmetrization.

For collinear spin-polarized calculations, RMG propagates the two spin channels as separate matrix problems. Each spin channel has its own $\mathbf{P}_{\mathbf{k}}^{\uparrow}$ or $\mathbf{P}_{\mathbf{k}}^{\downarrow}$ and its own spin-dependent Hamiltonian. The implementation does not introduce off-diagonal spinor blocks, spin flips, transverse spin dynamics, or spin-orbit induced spin mixing in the collinear RT-TDDFT path used here. The channels are nevertheless coupled self-consistently: after the $\mathbf{k}$ -weighted density reconstruction, the spin density available on a given spin partition is combined with the opposite-spin density before the Hartree and exchange–correlation potentials are evaluated. The Hartree update depends on the total density, while the exchange–correlation update uses the spin-dependent functional form with both spin densities.

Current-like observables are accumulated separately for each spin channel in the implementation. In the notation of Secs. 2.4 and 2.5, the physical total charge-current response

is obtained from the spin-channel contributions,

$$\Delta J_\alpha(t) = \Delta J_\alpha^\uparrow(t) + \Delta J_\alpha^\downarrow(t). \quad (89)$$

The separate spin-channel traces can also be retained for spin-resolved analysis within the fixed collinear basis. This organization is used later to analyze the spin-up and spin-down optical channels of the NV^- center.

The parallel decomposition combines the real-space domain decomposition inherited from ground-state RMG with additional parallelism over $\mathbf{k}$ points, spin channels, and dense active-space matrices. Real-space operations such as density reconstruction, application of local potentials, and grid integrations are distributed over the real-space grid. The active-space Hamiltonian and density matrices are distributed using the matrix backend selected for the run, including ScaLAPACK and tiled matrix-multiplication paths. The refactored `rmg::tddft<OrbitalType, MatrixType>` class provides CPU and GPU execution paths through templated orbital and matrix types.

This organization exposes independent work over $\mathbf{k}$ points and spin channels while retaining the locality of the real-space RMG operations and the efficiency of distributed dense matrix algebra. It is this combination of real-space decomposition, $\mathbf{k}$ /spin decomposition, matrix parallelism, and accelerator support that makes the periodic density-matrix RT-TDDFT implementation applicable to large supercells, including the 4095-atom NV^- -center calculations discussed below.

3.4 Optical Spectra and Implementation-Specific Processing

The formal response relations in Sec. 2.5 define the optical conductivity and dielectric tensor in terms of the physical macroscopic charge-current density. RMG writes spin-resolved raw current-like traces, for which the available current-postprocessing workflow produces optical spectra in arbitrary units.

Using the raw current-evaluation matrix defined in Sec. 3.2, the quantity written by RMG for spin channel σ can be represented schematically as

$$R_\alpha^\sigma(t_n) = \mathcal{S}_\alpha \left\{ \sum_{\mathbf{k}} w_{\mathbf{k}} \text{Re Tr} [\mathbf{P}_{\mathbf{k}}^\sigma(t_n) \mathbf{M}_{\mathbf{k},\alpha}^{\text{raw},\sigma}] \right\}, \quad (90)$$

where $\mathcal{S}$ denotes the vector symmetrization applied after the parallel reductions. The raw trace includes $\mathbf{k}$ -point weights and the reduction over distributed $\mathbf{k}$ -point and matrix partitions. It is written separately for each spin channel and does not explicitly include multiplication by q_e , division by V_{cell} , or the sum over spin channels.

Consequently, $R_\alpha^\sigma(t)$ should not be identified directly with the physical macroscopic charge-current density $J_\alpha^\sigma(t)$ of Sec. 2.2. If the raw matrix is interpreted as a zero-field velocity-like matrix in atomic units, conversion to the formal current density would require, at minimum, the electron-charge factor, the cell-volume normalization, the spin summation

for the total current, and the corresponding unit conversion:

$$J_{\alpha}(t) \sim \frac{q_e}{V_{\text{cell}}} \sum_{\sigma=\uparrow,\downarrow} R_{\alpha}^{\sigma}(t), \quad (91)$$

with $q_e = -1$ in the internal atomic-unit convention. This is a schematic relation identifying missing normalization factors, not an operation currently applied by the RMG current-output routine. The code also prints a ground-state current as a diagnostic header entry, but this value is not directly subtracted from the written time trace. In a fresh calculation, the first numerical current row is written after the first accepted propagation step, using the step-counter time label.

For current-based spectral analysis, the available RMG workflow first removes a numerical baseline from the sampled trace. For current files, it then applies a cumulative summation of the current-like signal, which acts as an implementation-specific discrete current-to-polarization integration. The resulting time series may be detrended and damped or windowed, Fourier transformed, normalized by the raw perturbation coefficient, and smoothed to produce the plotted spectrum. This processing is useful for comparing spectral shapes and peak positions, but the available workflow does not preserve the full signed complex phase of the response. It therefore does not by itself produce a quantitatively normalized complex conductivity or dielectric function.

The spectra presented below are reported in arbitrary units and are interpreted primarily through relative peak positions, spectral shape, polarization dependence, spin dependence, and active-space convergence rather than absolute optical intensities. A fully normalized conductivity or dielectric response can be obtained from the formal relations of Sec. 2.5, but doing so requires retaining the complex Fourier response and applying the physical charge, volume, spin-summation, field-normalization, and unit-conversion factors explicitly.

4 Validation of Periodic and Spin-Polarized RT-TDDFT

We first validate the periodic and spin-polarized RT-TDDFT implementation in RMG by comparing current-derived optical spectra with independent real-time calculations performed with `Octopus`, an established real-space RT-TDDFT code for finite and extended systems.¹³ The benchmark set was chosen as a progression of increasing complexity: a one-dimensional periodic hydrogen-dimer reference, pristine monolayer hBN, bulk Si, and spin-polarized CrI₃. The purpose of these comparisons is not to establish quantitatively normalized optical conductivities or dielectric functions, but to test whether RMG reproduces the principal spectral features obtained from an independent RT-TDDFT implementation at a comparable semilocal-DFT level. Accordingly, Fig. 1 reports optical spectra in arbitrary units and the comparison emphasizes peak positions, dominant spectral structures, overall spectral envelopes, and polarization or spin dependence where applicable.

The one-dimensional hydrogen-dimer system provides the simplest periodic reference. The model consists of hydrogen dimers repeated along one lattice direction with large transverse vacuum, and therefore tests the basic k-resolved propagation and current-response machinery without the additional complications of a multidimensional crystal. RMG repro-

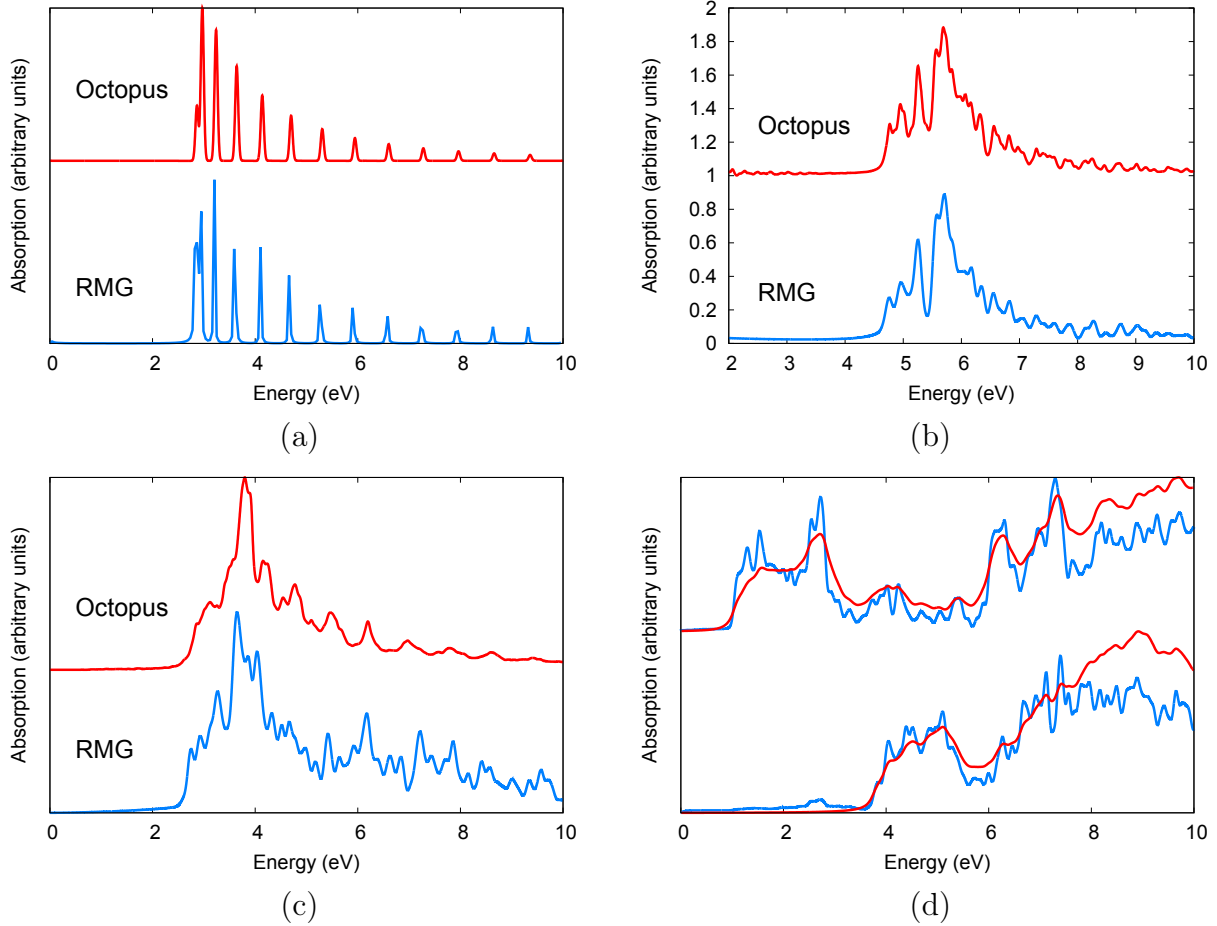


Figure 1: Optical spectra in arbitrary units from periodic RT-TDDFT calculations with RMG and independent `Octopus` reference calculations for systems of increasing complexity: (a) a one-dimensional periodic hydrogen-dimer reference, (b) pristine monolayer hBN, (c) bulk Si, and (d) spin-polarized CrI_3 . The comparison emphasizes peak positions and dominant spectral features rather than absolute intensity.

duces the sequence of major peaks obtained from the `Octopus` calculation over the plotted energy range, providing a basic cross-code validation of the periodic propagation and current-response implementation.

Pristine monolayer hBN provides a genuinely two-dimensional periodic benchmark. The RMG calculation uses a non-spin-polarized monolayer slab with vacuum normal to the layer and dense in-plane k-point sampling. Similar monolayer hBN tests have been used in recent periodic velocity-gauge RT-TDDFT implementation work.¹¹ The RMG and `Octopus` spectra show a similar low-energy onset and similar dominant structures in the main spectral envelope, validating the two-dimensional periodic implementation and the in-plane current response for an insulating material. As expected for semilocal PBE-level calculations, this benchmark should not be interpreted as a quantitative description of the strongly excitonic experimental optical spectrum of low-dimensional BN materials, where long-range electron-hole interactions are known to be important.²⁹ Detailed spin-polarized RT-TDDFT applications to hBN color centers are treated separately in a companion study.

Bulk Si then tests the full three-dimensional periodic implementation, including dense

k-point sampling and the complex matrix path required away from Γ . The spectra from RMG and `Octopus` show similar low-energy behavior and reproduce the dominant structure in the approximately 3–4 eV region, while somewhat larger differences are visible in the higher-energy fine structure. This level of agreement is consistent with the role of Si as a standard benchmark for periodic velocity-gauge RT-TDDFT implementations.¹⁸

Finally, CrI_3 provides the most demanding member of the general benchmark set: a spin-polarized periodic calculation including DFT+U for the Cr 3d states. The available RMG input is collinear spin-polarized and includes a $U = 3.0$ eV correction for the Cr 3d states. The comparison in Fig. 1 shows that RMG captures the main spectral structures obtained in the independent calculation for this magnetic material. We interpret this benchmark as a test of the spin-polarized periodic RT-TDDFT workflow, not as a complete treatment of spin-orbit-coupled or excitonic magneto-optical physics in CrI_3 , which requires physics beyond the present collinear PBE+U-level workflow.³⁰

Taken together, these four benchmarks show that the RMG implementation reproduces the principal spectral features of an independent RT-TDDFT code across one-dimensional, two-dimensional, three-dimensional, and spin-polarized periodic systems. The following NV^- -center calculations provide an additional physics-specific validation layer by testing defect orbital character, polarization selection rules, spin-channel selectivity, active-space convergence, and supercell-size effects in large diamond supercells.

5 Large-Scale Demonstration: NV^- Centers in Diamond

As a final scientific application, we apply the periodic collinear spin-polarized RT-TDDFT implementation to negatively charged nitrogen-vacancy centers (NV^-) in diamond. The NV^- center is one of the most widely studied solid-state point defects because its spin-triplet ground state is optically addressable and its long coherence times make it useful for quantum sensing, single-photon emission, and quantum information applications.^{24,25,31–37} For the present work, the NV^- center also provides a stringent large-scale test of the method: it combines localized defect states, spin-polarized electronic structure, polarization-resolved optical selection rules, active-space convergence requirements, and long-range periodic image effects in a three-dimensional host.^{25,31,38}

The goal of this section is not to provide a complete quantitative description of NV^- photophysics. The experimental zero-phonon line (ZPL) involves excited-state relaxation, vibronic coupling, and electron–phonon effects that are outside the fixed-ion adiabatic-PBE response considered here.^{24,25} Instead, the experimental $E_{\text{ZPL}} = 1.945$ eV is used as a reference energy for the defect-associated optical response.^{24,39} We first establish that the calculation reproduces the expected local defect orbital character, polarization selection rule, and spin-channel selectivity. We then examine active-space dependence and use a common 4096-site diamond framework to study how the optical spectrum evolves from concentrated periodic NV^- arrays toward the dilute-defect limit.

All NV^- calculations reported here used RMG with periodic boundary conditions, the PBE exchange-correlation functional, and the `NC_accuracy` norm-conserving pseudopotentials. The bulk diamond lattice constant was optimized with the same functional and pseudopotentials, giving $a_{\text{diamond}} = 3.5725125$ Å, and was then held fixed for the defect calculations.

The negatively charged defect cells were treated with a uniform compensating background charge.

5.1 Defect Electronic Structure and Optical Selection Rules

The NV^- center consists of a substitutional nitrogen atom adjacent to a carbon vacancy, with the defect axis oriented along the crystallographic $[111]$ direction. This axis defines the C_3 axis of the approximately C_{3v} defect. The defect introduces localized states in the diamond band gap. In the standard single-particle description, the triplet ground state associated with the 3A_2 manifold has the occupation $a_1^2 e_x^1 e_y^1$, with the a_1 -like orbital and the two components of the e manifold localized primarily on the nitrogen and the three carbon atoms adjacent to the vacancy.^{24,25,31}

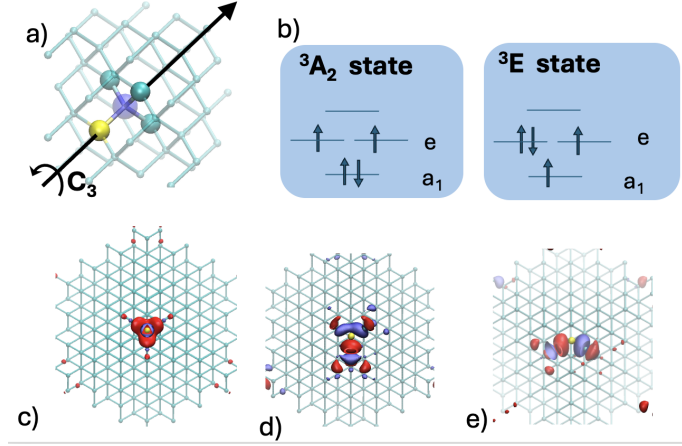


Figure 2: Electronic structure of the negatively charged NV^- center in diamond. (a) Local atomic structure of the defect, consisting of a substitutional nitrogen atom adjacent to a carbon vacancy. The arrow indicates the $[111]$ direction, which defines the C_3 axis of the approximately C_{3v} defect. (b) Schematic occupation of the defect levels in the triplet ground-state configuration, $a_1^2 e_x^1 e_y^1$, and in the defect-associated optically excited triplet configuration. The experimental 1.945 eV ZPL is used in this work only as a reference energy for the defect-region response. (c–e) Representative real-space isosurfaces of Kohn–Sham defect orbitals localized in the diamond band gap: the occupied a_1 -like orbital and the two components of the nearly degenerate e manifold.

The optical selection rule for the defect-associated ${}^3A_2 \rightarrow {}^3E$ transition provides a direct physics-based test of the periodic spin-polarized RT-TDDFT calculation.^{25,40} For the NV^- center, the vector component parallel to the defect axis transforms as A_1 , whereas the two transverse components transform as the two-dimensional E representation.^{24,41} Thus an axial perturbation does not produce an allowed electric-dipole transition between the 3A_2 and 3E manifolds,

$$A_2 \otimes A_1 \otimes E = E, \quad (92)$$

while a transverse perturbation is symmetry allowed,

$$A_2 \otimes E \otimes E = A_1 \oplus A_2 \oplus E. \quad (93)$$

We tested this selection rule using the 512-site NV- $x4$ supercell containing 511 atoms after defect formation. The spectra in Fig. 3 compare an impulsive perturbation parallel to the defect axis,

$$\hat{\mathbf{e}}_{\parallel} = \frac{1}{\sqrt{3}}(1, 1, 1), \quad (94)$$

with a transverse perturbation,

$$\hat{\mathbf{e}}_{\perp} = \frac{1}{\sqrt{2}}(1, -1, 0). \quad (95)$$

The defect-associated feature near the experimental ZPL reference is strongly activated for transverse polarization and strongly suppressed for axial polarization, consistent with the expected C_{3v} optical selection rule.

The spin-resolved response further identifies the active local spin channel. In the global collinear convention used in Fig. 3, the transverse defect-region feature is dominated by the spin-down channel. This corresponds to the local minority-spin $a_1 \rightarrow e$ excitation in the triplet defect configuration.^{25,40} Because the optical perturbation is spin independent, this should not be interpreted as a spin-flip transition; rather, it shows that the spin-conserving response is localized in the spin channel containing the relevant occupied defect orbital.

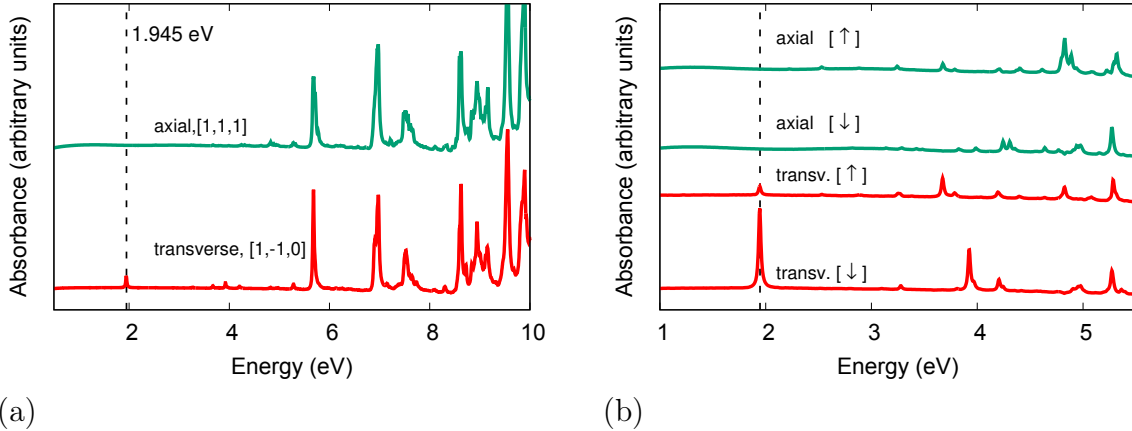


Figure 3: Polarization and spin selectivity of the NV⁻ defect optical response from spin-collinear RT-TDDFT for the 512-site NV- $x4$ supercell. The spectra are optical spectra in arbitrary units and are vertically shifted for clarity. (a) Spin-summed response for a perturbation parallel to the NV axis, $[111]$, and for a transverse perturbation along $[1\bar{1}0]$. The dashed vertical line marks the experimental NV⁻ ZPL energy, $E_{\text{ZPL}} = 1.945$ eV, as a reference. The defect-associated feature near this energy is strongly enhanced for transverse polarization and suppressed for axial polarization. (b) Expanded spin-resolved view of the sub-gap response. In the global spin convention used here, the transverse response near the ZPL reference is dominated by the spin-down channel, consistent with a local minority-spin $a_1 \rightarrow e$ excitation.

5.2 Active-Space Dependence of the Optical Response

The finite density-matrix propagation described in Sec. 3.1 requires selecting an active subspace of ground-state Kohn–Sham orbitals. This choice is especially important for large defect supercells, where including all occupied and a large number of unoccupied states can become prohibitively expensive. We therefore examined the active-space dependence of the

defect-associated optical response using the same NV- $x4$ supercell and transverse polarization as in Fig. 3.

The NV- $x4$ ground-state calculation has 1024 occupied states. The reference active space retained all occupied states and included 3200 unoccupied states, corresponding to 6.25 virtual states per pristine diamond site in the 512-site cell. The virtual-space tests used 512, 1024, 2048, and 3200 propagated unoccupied states, corresponding to 1, 2, 4, and 6.25 virtual states per site. The occupied-space tests varied the RMG starting-state index, `tddft_start_state`, over the values 0, 200, 400, 512, 600, and 800. Since this index defines the lower boundary of the propagated orbital subspace, the corresponding numbers of retained occupied states are 1024, 824, 624, 512, 424, and 224.

Figure 4 shows that the defect-region feature is sensitive to both parts of the active manifold. Increasing the number of propagated unoccupied states produces a systematic redshift of the defect-associated feature that gradually saturates as the active space is enlarged, indicating convergence with respect to the virtual-state manifold over the tested range. Truncating the occupied manifold shifts the response to higher energy, with the largest change occurring for the most aggressive occupied-space truncation. These results show that the localized defect excitation is not fully described by a minimal defect-only active space; the surrounding diamond environment and the response of the occupied manifold contribute measurably to the calculated optical feature.

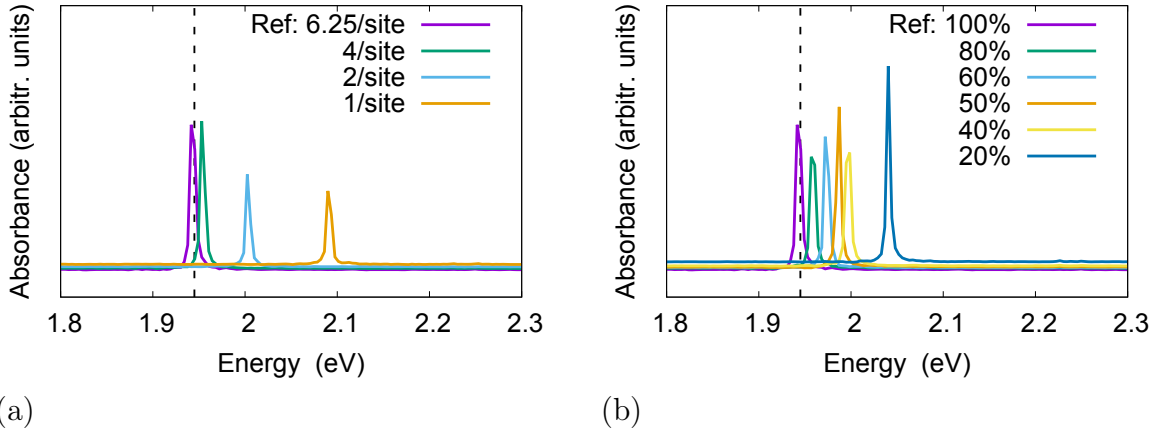


Figure 4: Active-space dependence of the defect-associated optical response for the 512-site NV- $x4$ supercell at the Γ point, using transverse polarization along $[1\bar{1}0]$. The spectra are optical spectra in arbitrary units. The dashed vertical line marks the experimental NV⁻ ZPL energy, $E_{\text{ZPL}} = 1.945$ eV, as a reference. The reference calculation in both panels uses `tddft_start_state` = 0, retains all 1024 occupied states, and includes 3200 unoccupied states. (a) Dependence on the number of virtual states per pristine diamond site, $n_{\text{virt}} = N_{\text{virt}}/N_{\text{site}}$, with $N_{\text{virt}} = 512, 1024, 2048$, and 3200. Increasing the virtual active space red-shifts the defect-region feature and approaches saturation over the tested range. (b) Dependence on occupied-space truncation. The starting-state indices 0, 200, 400, 512, 600, and 800 correspond to retaining 1024, 824, 624, 512, 424, and 224 occupied states, respectively. Aggressive occupied-space truncation produces a measurable blueshift.

This active-space behavior motivated the design of the final large-supercell concentration study below. To avoid conflating finite-size effects with changes in the propagated active space, the final concentration comparison was performed within a common 4096-site diamond

framework using the same active-space construction.

5.3 Defect Concentration and the Large-Supercell Limit

To isolate the effect of NV–NV separation from changes in overall cell size, we constructed a concentration series within the same nominal 4096-site diamond framework. The three systems contain 64, 8, and 1 NV[−] centers, respectively. The 64-NV system forms a regular $4 \times 4 \times 4$ simple-cubic lattice of NV-center nitrogen positions embedded in the diamond host, with a nearest-neighbor NV–NV spacing of approximately 7.15 Å, or $2a_{\text{diamond}}$. The 8-NV system is obtained by taking every other site of this defect lattice, giving a $2 \times 2 \times 2$ simple-cubic NV array with spacing approximately 14.29 Å, or $4a_{\text{diamond}}$. The single-defect periodic system has one NV[−] center in the 4096-site host, giving 4095 actual atoms after replacing one carbon by nitrogen and removing a nearest-neighbor carbon atom. Its periodic image separation is approximately 28.58 Å, or $8a_{\text{diamond}}$.

All three final concentration calculations used the same starting-state index of 2700 and 8200 unoccupied states. Because the systems contain different numbers of electrons, the total propagated active-space dimension is not exactly identical in every case, but these choices produce comparable propagated subspaces with dimensions of approximately 1.37×10^4 states. The 8200 unoccupied states correspond to roughly two virtual states per pristine diamond site. The RT-TDDFT trajectories were propagated for at least 42000 time steps with $\Delta t = 1.0$ a.u., corresponding to approximately 1 ps of real-time propagation.

Figure 5 compares the self-consistent RT-TDDFT spectra with independent-particle approximation (IPA) spectra computed using the LOPTICS implementation in RMG. For each corresponding concentration case, the IPA and RT-TDDFT calculations use the same geometry, magnetic configuration, k -point setup, and ground-state electronic structure. This provides a direct comparison between the independent-particle and self-consistent real-time responses for the same underlying electronic structure. The IPA spectra expose the underlying Kohn–Sham transition structure, while the RT-TDDFT spectra include the induced time-dependent Hartree and adiabatic exchange-correlation response. The IPA spectra are not treated as an exact reference.

The concentration dependence is pronounced. For the 64-NV array, the defect-region spectral weight is redistributed over several nearby features around the experimental ZPL reference. This fragmentation is consistent with stronger NV–NV coupling at the higher defect concentration, where the characteristic separation is only ~ 7.15 Å. At the 8-NV concentration, where the NV–NV spacing is doubled, the defect-region response is less fragmented. In the single-NV 4095-atom calculation, where the periodic image separation is approximately 28.58 Å, the defect-region response becomes substantially simpler and approaches the dilute-defect limit. The comparison demonstrates that large real-space cells are needed to separate the intrinsic local defect response from the finite-concentration spectral structure of periodic NV arrays.

5.4 Effect of Collinear Magnetic Configuration

The fixed-volume NV arrays also allow us to examine how the global collinear magnetic configuration affects the spin-resolved optical response. We considered parallel FM and G-

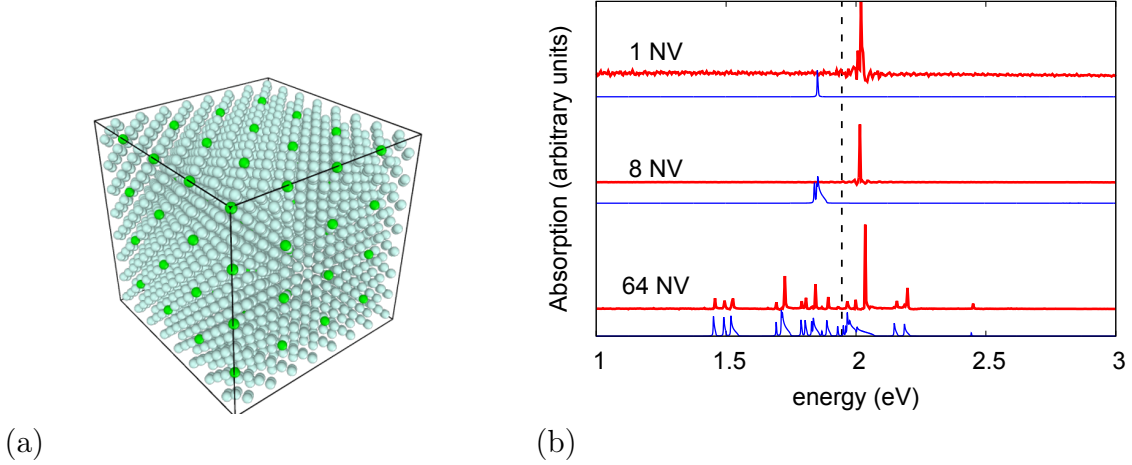


Figure 5: Defect-concentration dependence of the NV^- optical response in a common 4096-site diamond framework. (a) Example structure for the 64-NV array, where the NV centers form a regular $4 \times 4 \times 4$ simple-cubic defect lattice embedded in diamond. (b) Optical spectra in arbitrary units for the 1-, 8-, and 64-NV systems. Red curves show spin-collinear RT-TDDFT spectra and blue curves show independent-particle spectra from LOPTICS. The spectra are displayed with vertical offsets for clarity. The dashed vertical line marks the experimental $E_{\text{ZPL}} = 1.945$ eV reference energy. At high NV concentration, the spectral weight is distributed among several nearby defect-region features. As the NV–NV separation increases from approximately 7.15 Å to 14.29 Å and finally to the single-NV periodic image separation of approximately 28.58 Å, the defect-region RT-TDDFT response becomes substantially simpler and approaches the dilute-defect limit.

type antiferromagnetic (G-AFM) broken-symmetry initial configurations for the 64- and 8-NV arrays in the 4096-site host. For the 64-NV system, the G-AFM configuration alternates the local spin orientation between neighboring sites of the effective $4 \times 4 \times 4$ simple-cubic NV lattice. The 8-NV system uses the analogous alternating pattern on the reduced $2 \times 2 \times 2$ defect lattice. These configurations are used as controlled collinear spin arrangements; no claim is made here that G-AFM is the equilibrium magnetic ground state of an NV array.

The interpretation follows from the local minority-spin selection rule identified for the single NV^- center. In the FM configuration, all NV centers have the same local spin orientation, so the optically active local minority-spin transition maps predominantly onto one global spin channel. Contributions from equivalent NV centers then accumulate in that same global channel. In the G-AFM configuration, neighboring defects have opposite local spin orientations. The local minority-spin transition therefore maps onto the spin-up global channel on one magnetic sublattice and the spin-down global channel on the other. As a result, each global spin channel receives contributions from approximately half of the NV centers.

This behavior is visible in Fig. 6. For a fixed defect concentration, the dominant FM response in its active global spin channel is roughly twice the response in either individual G-AFM global spin channel, consistent with the redistribution expected from the two magnetic sublattices. This is a qualitative-to-semiquantitative interpretation of the plotted spectra, whose curves were processed in the same way for the FM and G-AFM cases. The result does not imply a spin-flip optical transition: the optical perturbation remains spin independent,

and the observed redistribution reflects the mapping between local minority spin and the fixed global collinear spin basis.

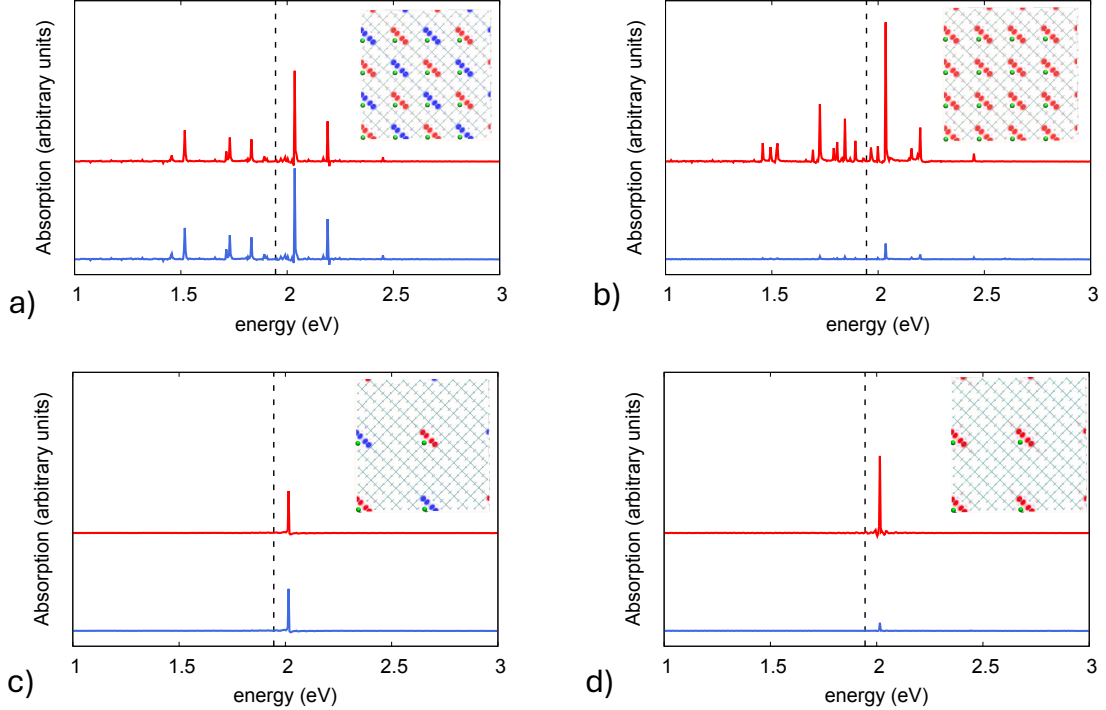


Figure 6: Spin-resolved RT-TDDFT optical spectra in arbitrary units for replicated NV^- arrays in the 4096-site diamond host. Blue curves denote the global spin-up response and red curves denote the global spin-down response. Panels (a) and (b) show the 64-NV system in G-AFM and FM collinear configurations, respectively; panels (c) and (d) show the corresponding 8-NV G-AFM and FM configurations. The insets show cross-sectional maps of the differential spin density and the initialized local moment pattern. All spectral curves in this figure were processed in the same way, so relative intensities between FM and G-AFM spectra at a fixed concentration can be compared. In the FM configurations, the local minority-spin defect excitation maps predominantly onto one global spin channel. In the G-AFM configurations, the same local minority-spin excitation is distributed between the two global spin channels because the local moment orientation alternates between the two magnetic sublattices.

Together, the NV^- calculations show that the periodic spin-polarized RT-TDDFT implementation captures the local optical selection rules and spin-channel structure of a well-characterized defect while scaling to a 4095-atom single-defect diamond calculation. The concentration series demonstrates how the defect-region optical spectrum evolves as NV^- coupling is reduced with decreasing defect concentration and the system approaches the dilute-defect limit, and the FM/G-AFM comparison shows how local spin selectivity is represented in global collinear spin channels for extended defect arrays.

References

- (1) Marques, M. A. L., Maitra, N. T., Nogueira, F. M. S., Gross, E. K. U., Rubio, A., Eds. *Fundamentals of Time-Dependent Density Functional Theory*; Lecture Notes in

Physics; Springer: Heidelberg, Germany, 2012; Vol. 837.

- (2) Li, X.; Govind, N.; Isborn, C.; A. Eugene DePrince, I.; ; Lopata, K. Real-Time Time-Dependent Electronic Structure Theory. *Chem. Rev.* **2020**, *120*, 9951 – 9993.
- (3) Yabana, K.; Bertsch, G. F. Time-dependent Local-density Approximation in Real Time. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1996**, *54*, 4484–4487.
- (4) Jakowski, J.; Irle, S.; Sumpter, B.; Morokuma, K. Modeling charge transfer in fullerene collisions via real-time electron dynamics. *J. Phys. Chem. Lett* **2012**, *3*, 1536–1542.
- (5) Lingerfelt, D. B.; Ganesh, P.; Jakowski, J.; Sumpter, B. G. Understanding Beam Induced Electronic Excitations in Materials. *J. Chem. Theory and Comput.* **2020**, *16*, 1200–1214.
- (6) Yu, T.; Lingerfelt, D.; Jakowski, J.; Javed, M. A.; Ganesh, P.; Sumpter, B. G. Electron-Beam-Induced Molecular Plasmon Excitation and Energy Transfer in Silver Molecular Nanowires. *J. Phys. Chem. A* **2021**, *125*, 74–87.
- (7) Resta, R. Macroscopic polarization in crystalline dielectrics: the geometric phase approach. *Reviews of Modern Physics* **1994**, *66*, 899–915.
- (8) King-Smith, R. D.; Vanderbilt, D. Theory of polarization of crystalline solids. *Physical Review B* **1993**, *47*, 1651–1654.
- (9) Ditter, E.; Mattiat, J.; Lubert, S. The position operator problem in periodic calculations with emphasis on theoretical spectroscopy. *Phys. Chem. Chem. Phys.* **2023**, *25*, 14672.
- (10) Mattiat, J.; Lubert, S. Comparison of Length, Velocity, and Symmetric Gauges for the Calculation of Absorption and Electric Circular Dichroism Spectra with Real-Time Time-Dependent Density Functional Theory. *J. Chem. Theory Comput.* **2022**, *18*, 5513 – 5526.
- (11) Hanasaki, K.; Ali, Z. A.; Choi, M.; Del Ben, M.; Wong, B. M. Implementation of real-time TDDFT for periodic systems in the open-source PySCF software package. *Journal of Computational Chemistry* **2023**, *44*, 980–987.
- (12) Onida, G.; Reining, L.; Rubio, A. Electronic excitations: density-functional versus many-body Green’s-function approaches. *Rev. Mod. Phys.* **2002**, *74*, 601–659.
- (13) Tancogne-Dejean, N.; Oliveira, M. J. T.; Andrade, X.; Appel, H.; Borca, C. H.; Le Breton, G.; Buchholz, F.; Castro, A.; Corni, S.; Correa, A. A. et al. Octopus, a computational framework for exploring light-driven phenomena and quantum dynamics in extended and finite systems. *The Journal of Chemical Physics* **2020**, *152*, 124119.
- (14) W.Draeger, E.; Andrade, X.; Gunnels, J. A.; Bhatele, A.; Schleife, A.; A.Correa, A. Massively parallel first-principles simulation of electron dynamics in materials. *Journal of Parallel and Distributed Computing* **2017**, *106*, 205–214.

- (15) Shepard, C.; Ruiyi Zhou, D. C. Y.; Yao, Y.; Kanai, Y. Simulating electronic excitation and dynamics with real-time propagation approach to TDDFT within plane-wave pseudopotential formulation. *J. Chem. Phys.* **2021**, *155*, 100901.
- (16) Kühne, T.; Iannuzzi, M.; Ben, M. D.; Rybkin, V. V.; Seewald, P.; Stein, F.; Laino, T.; Khaliullin, R. Z.; Schütt, O.; Schiffmann, F. et al. CP2K: An electronic structure and molecular dynamics software package - Quickstep: Efficient and accurate electronic structure calculations. *J. Chem. Phys.* **2020**, *150*, 194103.
- (17) Pela, R. R.; Draxl, C. All-electron full-potential implementation of real-time TDDFT in exciting. *Electron. Struct.* **2021**, *3*, 037001.
- (18) Hekele, J.; Yao, Y.; Kanai, Y.; Blum, V.; Kratzer, P. All-electron real-time and imaginary-time time-dependent density functional theory within a numeric atom-centered basis function framework. *The Journal of Chemical Physics* **2021**, *155*, 154801.
- (19) Rufus, N. D.; Kanungo, B.; Gavini, V. Fast and robust all-electron density functional theory calculations in solids using orthogonalized enriched finite elements. *Phys. Rev. B* **2021**, *104*, 085112.
- (20) Xu, Q.; Del Ben, M.; Okay, M. S.; Choi, M.; Ibrahim, K. Z.; Wong, B. M. Velocity-Gauge Real-Time Time-Dependent Density Functional Tight-Binding for Large-Scale Condensed Matter Systems. *Journal of Chemical Theory and Computation* **2023**, *19*, 7989–7997.
- (21) Hourahine, B.; Aradi, B.; V. Blum, e. a. DFTB+, a software package for efficient approximate density functional theory based atomistic simulations. *J. Chem. Phys.* **2020**, *152*, 124101.
- (22) Jakowski, J.; Lu, W.; Briggs, E.; Lingerfelt, D.; Sumpter, B. G.; Ganesh, P.; Bernholc, J. Simulation of 24,000 Electron Dynamics: Real-Time Time-Dependent Density Functional Theory (TDDFT) with the Real-Space Multigrids (RMG). *Journal of Chemical Theory and Computation* **2025**, *21*, 1322–1339.
- (23) Briggs, E. L.; Lu, W.; Bernholc, J. Adaptive finite differencing in high accuracy electronic structure calculations. *npj Comput. Mater.* **2024**, *10*, 17.
- (24) Doherty, M. W.; Manson, N. B.; Delaney, P.; Jelezko, F.; Wrachtrup, J.; Hollenberg, L. C. L. The nitrogen-vacancy colour centre in diamond. *Phys. Rep.* **2013**, *528*, 1–45.
- (25) Gali, Á. Ab initio theory of the nitrogen-vacancy center in diamond. *Nanophotonics* **2019**, *8*, 1907–1943.
- (26) Kleinman, L.; Bylander, D. M. Efficacious Form for Model Pseudopotentials. *Phys. Rev. Lett.* **1982**, *48*, 1425.
- (27) Jakowski, J.; Morokuma, K. Liouville-von Neumann molecular dynamics. *J. Chem. Phys.* **2009**, *130*, 224106.

- (28) Magnus, W. On the exponential solution of differential equations for a linear operator. *Commun. Pure Appl. Math.* **1954**, *7*, 649.
- (29) Wirtz, L.; Marini, A.; Rubio, A. Excitons in Boron Nitride Nanotubes: Dimensionality Effects. *Physical Review Letters* **2006**, *96*, 126104.
- (30) Wu, M.; Li, Z.; Cao, T.; Louie, S. G. Physical origin of giant excitonic and magneto-optical responses in two-dimensional ferromagnetic insulators. *Nature Communications* **2019**, *10*, 2371.
- (31) Gali, Á.; Fyta, M.; Kaxiras, E. Ab initio supercell calculations on nitrogen-vacancy center in diamond: Electronic structure and hyperfine tensors. *Phys. Rev. B* **2008**, *77*, 155206.
- (32) Balasubramanian, G.; Neumann, P.; Twitchen, D.; Markham, M.; Kolesov, R.; Mizuochi, N.; Isoya, J.; Achard, J.; Beck, J.; Tissler, J. et al. Ultralong spin coherence time in isotopically engineered diamond. *Nat. Mater.* **2009**, *8*, 383–387.
- (33) Degen, C. L.; Reinhard, F.; Cappellaro, P. Quantum sensing. *Rev. Mod. Phys.* **2017**, *89*, 035002.
- (34) Maze, J. R.; Stanwix, P. L.; Hodges, J. S.; Hong, S.; Taylor, J. M.; Cappellaro, P.; Jiang, L.; Gurudev Dutt, M. V.; Togan, E.; Zibrov, A. S. et al. Nanoscale magnetic sensing with an individual electronic spin in diamond. *Nature* **2008**, *455*, 644–647.
- (35) Schirhagl, R.; Chang, K.; Loretz, M.; Degen, C. L. Nitrogen-vacancy centers in diamond: nanoscale sensors for physics and biology. *Annu. Rev. Phys. Chem.* **2014**, *65*, 83–105.
- (36) Aharonovich, I.; Englund, D.; Toth, M. Solid-state single-photon emitters. *Nat. Photonics* **2016**, *10*, 631–641.
- (37) Bradley, C. E.; Randall, J.; Abobeih, M. H.; Berrevoets, R. C.; Degen, M. J.; Bakker, M. A.; Markham, M.; Twitchen, D. J.; Taminiau, T. H. A ten-qubit solid-state spin register with quantum memory up to one minute. *Phys. Rev. X* **2019**, *9*, 031045.
- (38) Choi, S.; Jain, M.; Louie, S. G. Mechanism for optical initialization of spin in NV[−] center in diamond. *Phys. Rev. B* **2012**, *86*, 041202(R).
- (39) Davies, G.; Hamer, M. F. Optical studies of the 1.945 eV vibronic band in diamond. *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* **1976**, *348*, 285–298.
- (40) Gali, A.; Janzén, E.; Deák, P.; Kresse, G.; Kaxiras, E. Theory of Spin-Conserving Excitation of the N-V[−] Center in Diamond. *Physical Review Letters* **2009**, *103*, 186404.
- (41) Maze, J. R.; Gali, A.; Togan, E.; Chu, Y.; Trifonov, A.; Kaxiras, E.; Lukin, M. D. Properties of nitrogen-vacancy centers in diamond: the group theoretic approach. *New Journal of Physics* **2011**, *13*, 025025.